

A MODIFIED KERNEL DENSITY ESTIMATOR FOR RANDOMLY RIGHT-CENSORED DATA

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Summary: A modification of the kernel density estimator for randomly right-censored data is proposed. The advantage of this estimator is that the data play a role in the degree of smoothing of the estimator while retaining the desirable features of the kernel estimator. Convergence in probability and almost sure convergence of the estimator are discussed. Examples are given which illustrate the differences between this modified estimator and the "fixed-bandwidth" kernel estimator for randomly right-censored data.

1. INTRODUCTION

Many authors have studied the properties and applications of kernel density estimators since the appearance of the important work of Rosenblatt (1956) and Parzen (1962) on nonparametric density estimation. Parzen (1962), Nadaraya (1965), and Van

Ryzin (1969) showed that the kernel estimator was a consistent or strongly consistent estimator of the unknown probability density function. A disadvantage of the kernel estimator is that the bandwidth sequence, which controls the degree of smoothing of the estimator about the observations, is chosen without regard to the data (Cover, 1972). Wagner (1975) considered a modification of the kernel density estimator which allowed the data to play a role in this smoothing.

All of the results mentioned above have been for the complete sample case. However, in many life testing situations, observations may be arbitrarily or randomly censored or truncated from the right (Lagakos, 1979). This occurs in medical trials when patients may enter treatment at different times and then either die or leave the study at various times before it is terminated. Also in industrial life testing, items may be removed from the study at random times for further analysis or for other reasons. Thus, it is of importance to obtain non-parametric estimates of the probability density function f^0 of the lifetime random variable based on such arbitrarily right-censored data. Földes, Rejtő and Winter (1981) discussed a kernel density estimator of f^0 for this censored case based on the product-limit estimator of the survival distribution (Kaplan and Meier, 1958). They obtained strong consistency properties of the estimator under various conditions on f^0 , such as f^0 is bounded and either is uniformly continuous or has a bounded derivative. McNichols and Padgett (1981) obtained the limiting expected value, variance, and mean squared error of this estimator assuming proportional hazards for the life distribution and the censoring distribution.

In this paper, we modify the kernel density estimator of Földes, Rejtő and Winter (1981) and McNichols and Padgett (1981) for arbitrarily right-censored data to allow the actual observations to play a role in the degree of smoothing of the estimator. This estimator reduces to that of Wagner (1975) when no censoring is present and $x \geq 0$.

In section 2, the notation is presented and the estimator is defined. The consistency of the estimator is investigated in section 3. Some examples using data from Kalbfleisch and Prentice (1980) and simulated randomly right-censored data from various Weibull distributions are presented in section 4 to illustrate the smoothing effect of the modified estimator as compared to the "fixed-bandwidth" kernel estimator of Földes, Rejtő, and Winter (1981) and McNichols and Padgett (1981).

2. NOTATION AND PRELIMINARIES

Let $X_1^0, \dots, X_n^0$ be the survival times of n items or individuals which are censored from the right by a sequence $U_1, \dots, U_n$ of random variables. Let F^0 be the continuous distribution function of the X_i^0 's which are assumed to be nonnegative and identically distributed random variables and let f^0 be a corresponding density. Also, assume that $U_1, \dots, U_n$ are independent identically distributed random variables, independent of $X_1^0, \dots, X_n^0$, with common unknown continuous distribution function H . It is desired to estimate the density function f^0 based on the randomly censored observations (X_i, Δ_i) , $i=1, \dots, n$, where

$$X_i = \min(X_i^0, U_i), \quad \Delta_i = \begin{cases} 1 & \text{if } X_i^0 \leq U_i \\ 0 & \text{if } X_i^0 > U_i \end{cases}$$

Thus, it is known which observations are times of failure or death and which ones are right-censored by the random variables $\{U_i\}$. Note that $\Delta_1, \dots, \Delta_n$ are independent. The distribution function of each X_i , $i=1, \dots, n$, denoted by F with density f , is given by $1 - F = (1 - F^0)(1 - H)$. Let $Z_1, \dots, Z_n$ denote the order statistics of $X_1, \dots, X_n$ and let $\Delta_1', \dots, \Delta_n'$ be the corresponding Δ -values. The observed value of (Z_i, Δ_i') will be denoted by (z_i, δ_i') for each i .

Let $\hat{P}_n(t)$ denote the product-limit estimator of $1-F^0$ as modified by Efron (1967), defined by

$$\hat{P}_n(t) = \begin{cases} 1, & 0 \leq t \leq z_1 \\ \prod_{i=1}^{k-1} \left(\frac{n-i}{n-i+1} \right)^{\delta_i^1}, & t \in (z_{k-1}, z_k], k=2, \dots, n \\ 0, & t > z_n. \end{cases}$$

Denote the product-limit estimate of $F^0(t)$ by $\hat{F}_n(t) = 1 - \hat{P}_n(t)$, and let s_j denote the jump of $\hat{P}_n$ at z_j , that is,

$$s_j = \begin{cases} 1 - \hat{P}_n(z_2), & j = 1 \\ \hat{P}_n(z_j) - \hat{P}_n(z_{j+1}), & j = 2, \dots, n-1 \\ \hat{P}_n(z_n), & j = n. \end{cases}$$

Note that $s_j = 0$ if and only if $\delta_j^1 = 0$ for $j < n$.

Based on the randomly right-censored sample (X_i, Δ_i) , $i=1, \dots, n$, consider the following estimator of $f^0(x)$:

$$\begin{aligned} f_n(x) &= \left[\int_{-\infty}^{\infty} \Gamma_n^{-1} K\left(\frac{x-t}{\Gamma_n}\right) d\hat{F}_n(t) \right] \cdot I_{[0, \infty)}(x) \\ (2.1) \quad &= \left[\Gamma_n^{-1} \sum_{j=1}^n s_j K\left(\frac{x-z_j}{\Gamma_n}\right) \right] \cdot I_{[0, \infty)}(x), \end{aligned}$$

where $I_A(\cdot)$ is the indicator function of the set A , $\Gamma_n \equiv \Gamma_n(X_1, \dots, X_n)$ is some function of $X_1, \dots, X_n$, and $K(t)$ is a suitable kernel. Also, for any distribution function G define $T_G = \sup\{x \in \mathbb{R} : G(x) < 1\}$.

3. MAIN RESULTS

In this section, the consistency properties of the modified kernel estimator (2.1) are investigated. The first result is the weak point-wise consistency of f_n .

Theorem 1. Suppose $H(T_{F0}) < 1$, K has bounded variation, $\lim_{|x| \rightarrow \infty} xK(x) = 0$, and Γ_n satisfies

- (a) $\Gamma_n \rightarrow 0$ in probability; and
 (3.1) (b) $n^{\frac{1}{2}}(\log \log n)^{-\frac{1}{2}}\Gamma_n \rightarrow \infty$ in probability.

Then $f_n(x) \rightarrow f^0(x)$ in probability at each x for which f^0 is continuous.

The result follows similar to the proof of Wagner's (1975) Theorem 3 using the fact that $n^{\frac{1}{2}}(\log \log n)^{-\frac{1}{2}} \sup_{y \in R} |\hat{F}_n(y) - F^0(y)| \xrightarrow{a.s.} 0$ as $n \rightarrow \infty$ (Földes and Rejtő, 1981).

One needs to know how to choose Γ_n so that (3.1) holds. Analogous to Wagner's (1975) discussion, one procedure is as follows: Let $k(n) = [n^\beta]$, where $0 < \beta < 1$. Let D_{jn} denote the distance from Z_j to its $k(n)$ -th nearest neighbour from $Z_1, \dots, Z_n$ (with Z_j omitted), $1 \leq j \leq n$. Let Γ_n be selected as D_{jn} with probability s_j , $j = 1, 2, \dots, n$. Thus, if the j -th ordered observation is uncensored or $j = n$, its corresponding D_{jn} will be selected with probability $s_j > 0$. Otherwise, D_{jn} will not be selected for Γ_n . Note that if the observations are all uncensored, that is, the sample is complete, D_{jn} will be selected with probability $1/n$ for all $1 \leq j \leq n$, which is the case of Wagner (1975).

Theorem 2. For the choice of Γ_n given above

- (a) $\Gamma_n \rightarrow 0$ in probability; and
 (b) $n^{\frac{1}{2}}(\log \log n)^{-\frac{1}{2}}\Gamma_n \rightarrow \infty$ in probability if $\beta > \frac{1}{2}$.

The proof of theorem 2 follows from the fact that $n^{1-\beta} n^{-\frac{1}{2}}(\log \log n)^{-\frac{1}{2}} \rightarrow 0$ as $n \rightarrow \infty$ if and only if $\beta > \frac{1}{2}$ and

the proof of Wagner's (1975) Theorem 2, letting f in his proof be the density function of the observed X_i , $i=1, \dots, n$, and not the survival density f^0 .

Note that the condition $\beta > \frac{1}{2}$ is the same condition required by Wagner. Also, as in the complete sample case, if convergence in (3.1) is strengthened to with probability one, then the conclusion of theorem 1 can be strengthened to with probability one. Other quantities which will work for the choice of Γ_n in theorem 2 are $\max(D_{1n}, \dots, D_{nn})$ and $\min(D_{1n}, \dots, D_{nn})$.

The extension of theorem 1 to the multivariate case is not considered here. The analog of the Kaplan-Meier estimator has been proposed for bivariate distributions by Campbell (1981), but the problem of kernel density estimation under random censorship in several dimensions is still under investigation.

4. EXAMPLES

Let $\hat{f}_n(x)$ denote the "fixed-bandwidth" kernel estimator discussed by McNichols and Padgett (1981). That is, $\{\Gamma_n\}$ in (2.1) is replaced by a nonrandom sequence $\{h_n\}$ satisfying the conditions of Parzen (1962). The optimal bandwidth derived by Parzen in the sense of minimum mean squared error is $h_n = O(n^{-1/5})$. In this section several examples are given. The first example uses the censored data of Kalbfleisch and Prentice (1980, p. 223, standard, squamous data set) to illustrate the smoothing effect of the modified estimator (2.1) as compared with the fixed bandwidth estimator with $h_n = n^{-1/5}$. Although this choice of h_n is not exactly optimal, even for the complete sample case, it gives a reasonably smooth estimate. Other choices of h_n such as $h_n = cn^{-1/5}$ behave in a similar manner, becoming very smooth for c as large as $c = 10$ and very rough with sharp peaks for c less than 0.5 for this data.

The ordered observations from Kalbfleisch and Prentice (1980) (transformed to "months") are:

i	1	2	3	4	5	6	7	8
z_i	0,27	0,33	0,37	0,83 ⁺	1,40	2,40	2,73	3,33 ⁺

i	9	10	11	12	13	14	15
z_i	3,67	3,93	4,20	4,80	7,60	10,47	13,70

The "+" denotes a censored observation, so that $\delta_4^1 = \delta_8^1 = 0$ and all other δ_i^1 are one. The estimates $f_n(x)$ and $\hat{f}_n(x)$ were computed from this data for $0 < x < 17$. The values of Γ_n were chosen as described in the paragraph preceding theorem 2.

Figure 1 shows the estimates. The fixed-bandwidth estimator has sharp peaks occurring at the uncensored observations (the last observation is always treated as uncensored in the modified product-limit estimator described in section 2), while $\hat{f}_n(x)$ is very smooth. The values of β used in the calculations for figure 1 are 0,52 and 0,60. The optimal choice of the value of the "smoothing parameter" β is still an open question. Analogous to the choice of h_n in the fixed bandwidth estimator, small values of β produce rougher estimates than values of β near one. However, large β values produce too much smoothing. For this example, the "moderate" values of β between 0,5 and 0,6 satisfying (b) of theorem 2 gave pleasing results.

For other examples illustrating the modified estimator, randomly right-censored samples of size $n = 60$ were generated from various Weibull distributions with shape parameter ξ and scale parameter α . The censoring values were generated also from a Weibull distribution with the same shape parameter ξ but different scale parameter γ . Figures 2-5 show some of the results. The true life distribution is shown with the fixed bandwidth kernel estimator $\hat{f}_n$ and the modified kernel estimator f_n with

$\beta = 0,52$. Again, Γ_n was chosen as in the paragraph preceding theorem 2. These examples indicate that although the modified estimator f_n is more difficult to compute, it generally produces "nicer" estimates than the fixed bandwidth estimator.

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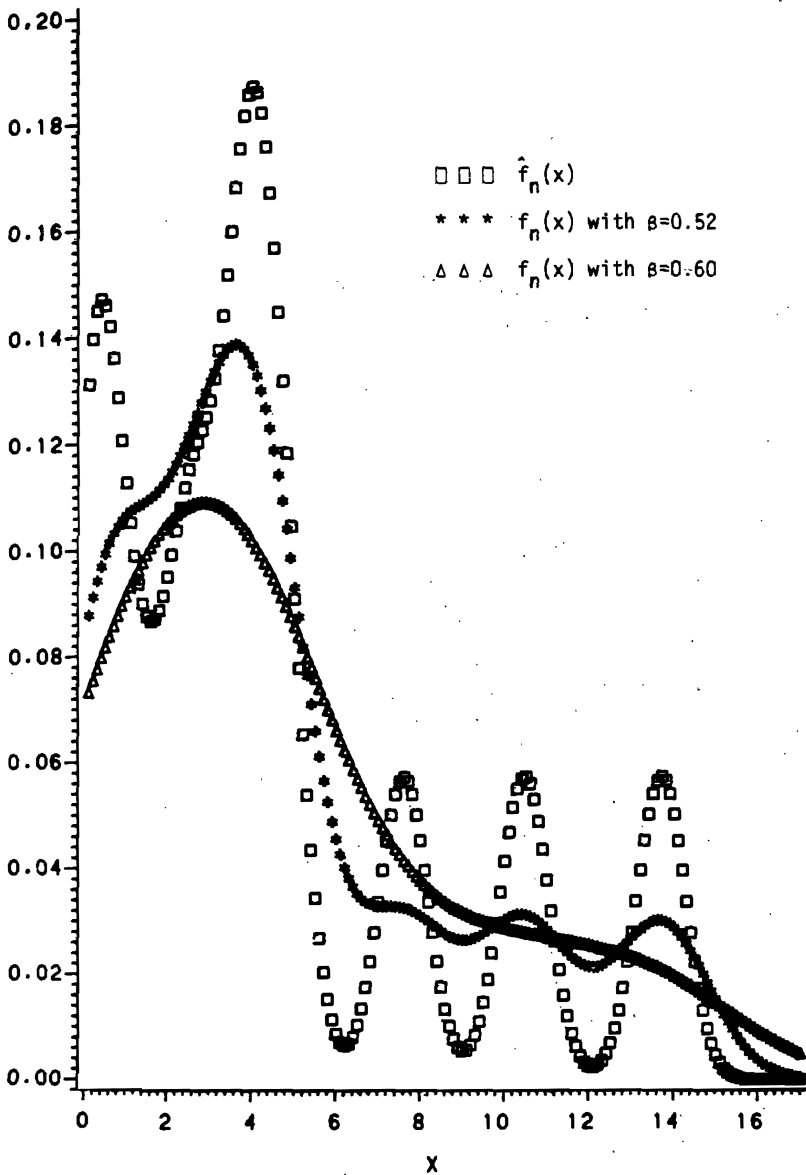


FIGURE 1. Density estimates from Kalbfleisch-Prentice data

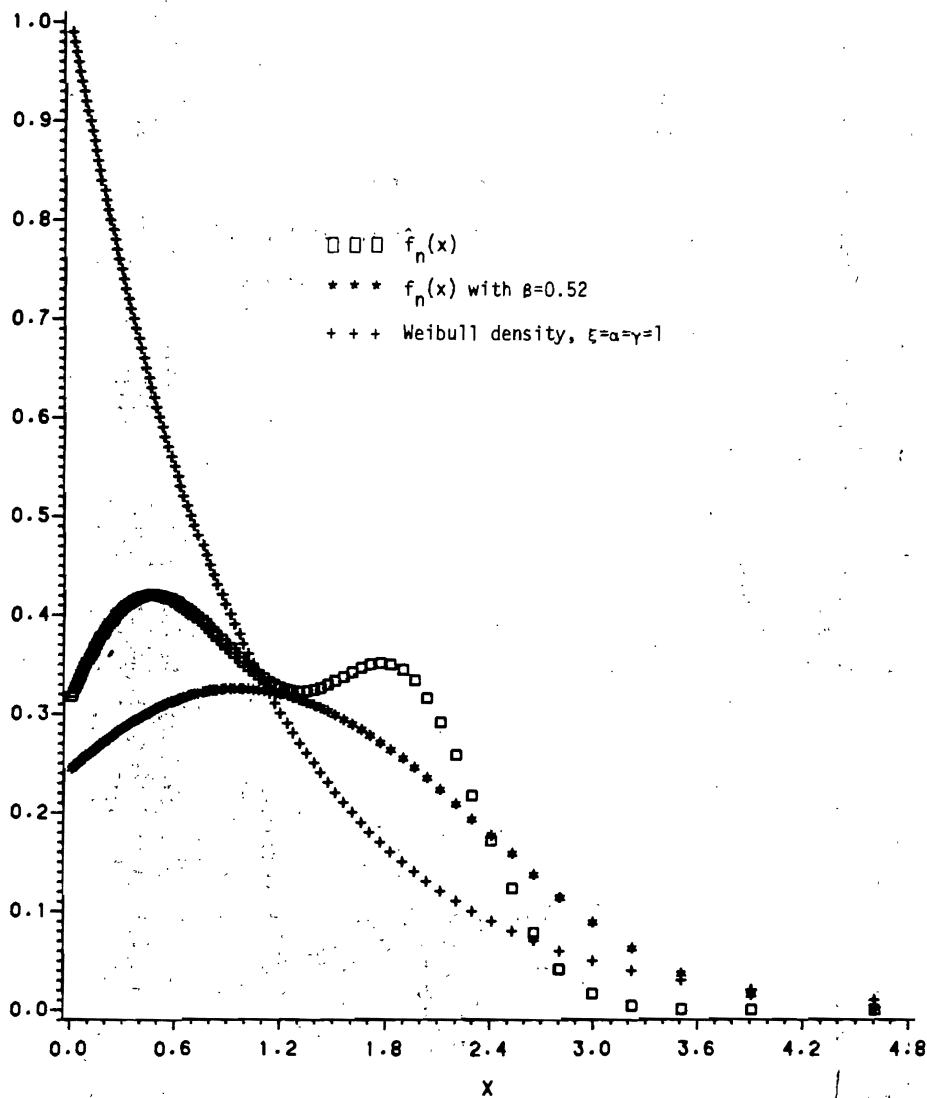


FIGURE 2. Density estimates from Weibull data for $n=60$

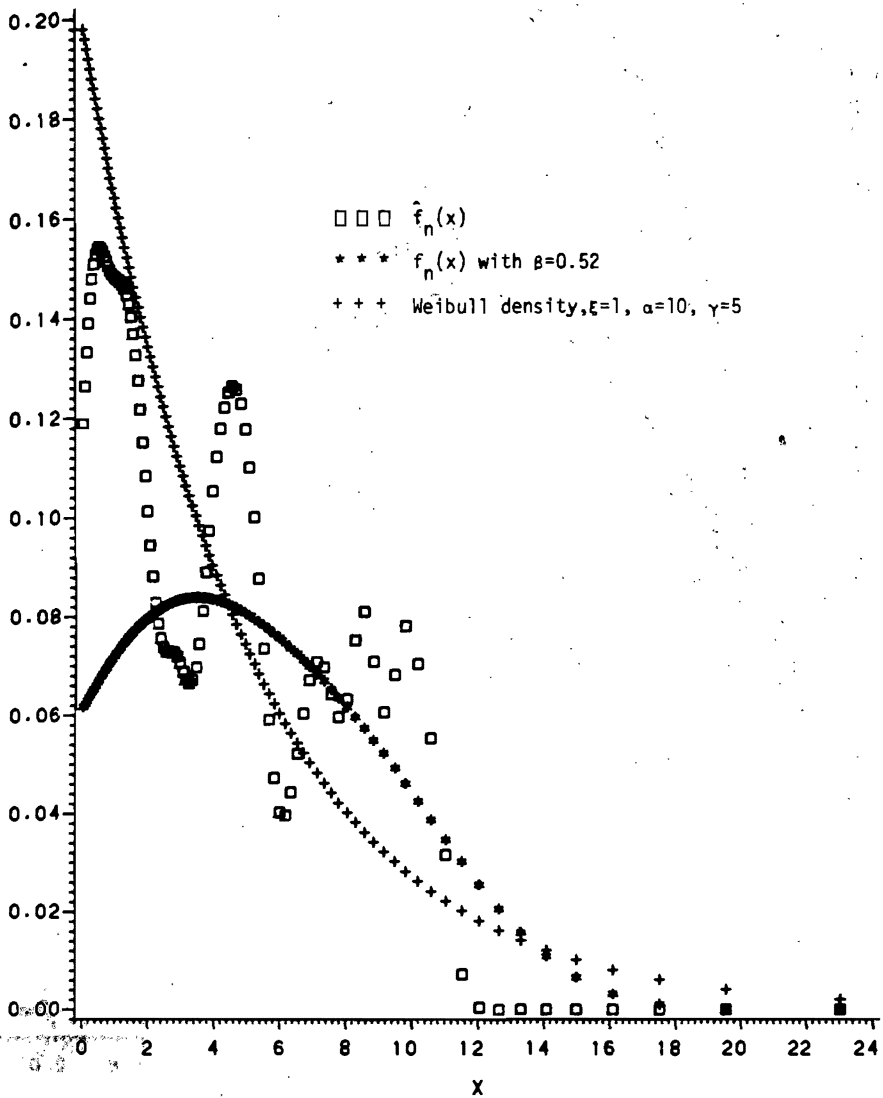


FIGURE 3. Density estimates from Weibull data for $n=60$

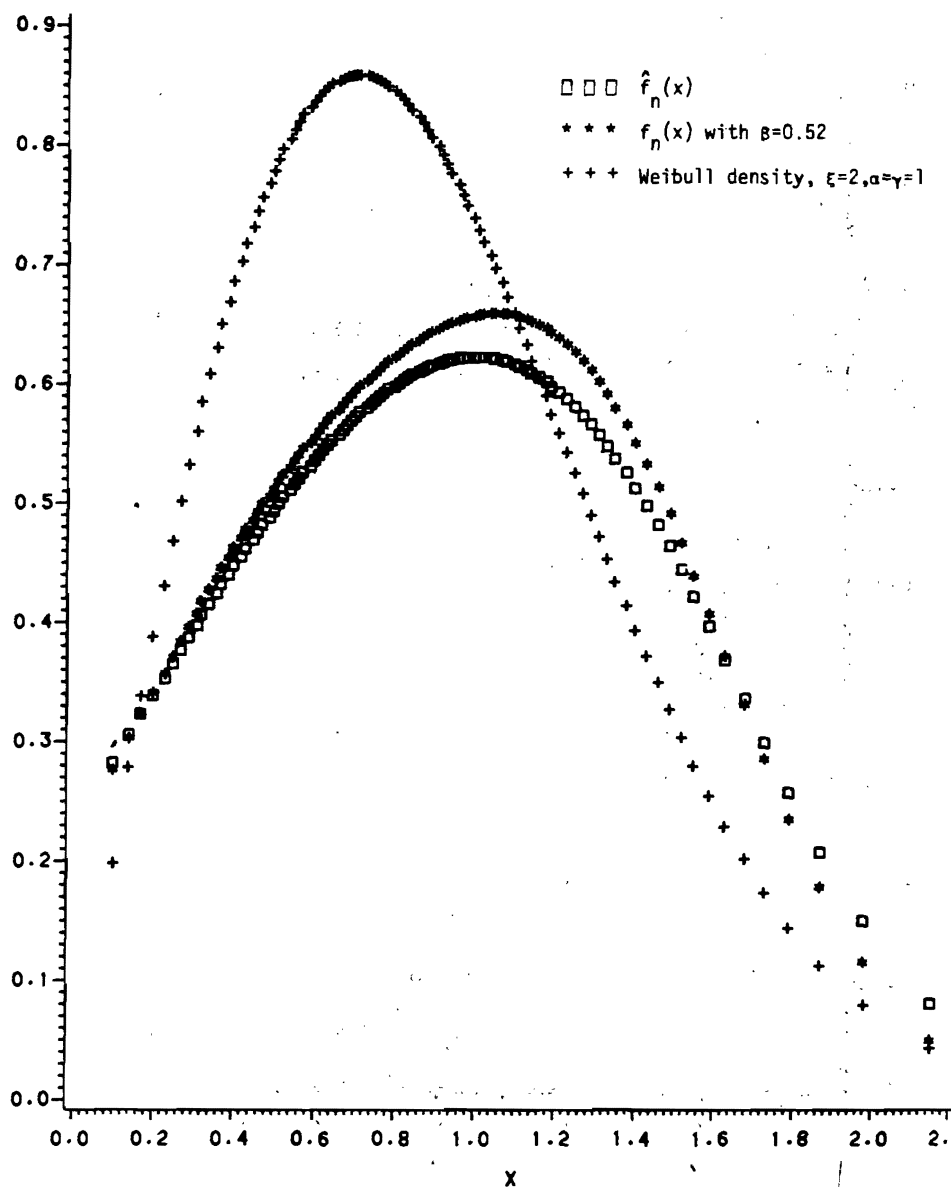


FIGURE 4. Density estimates from Weibull data for $n=60$

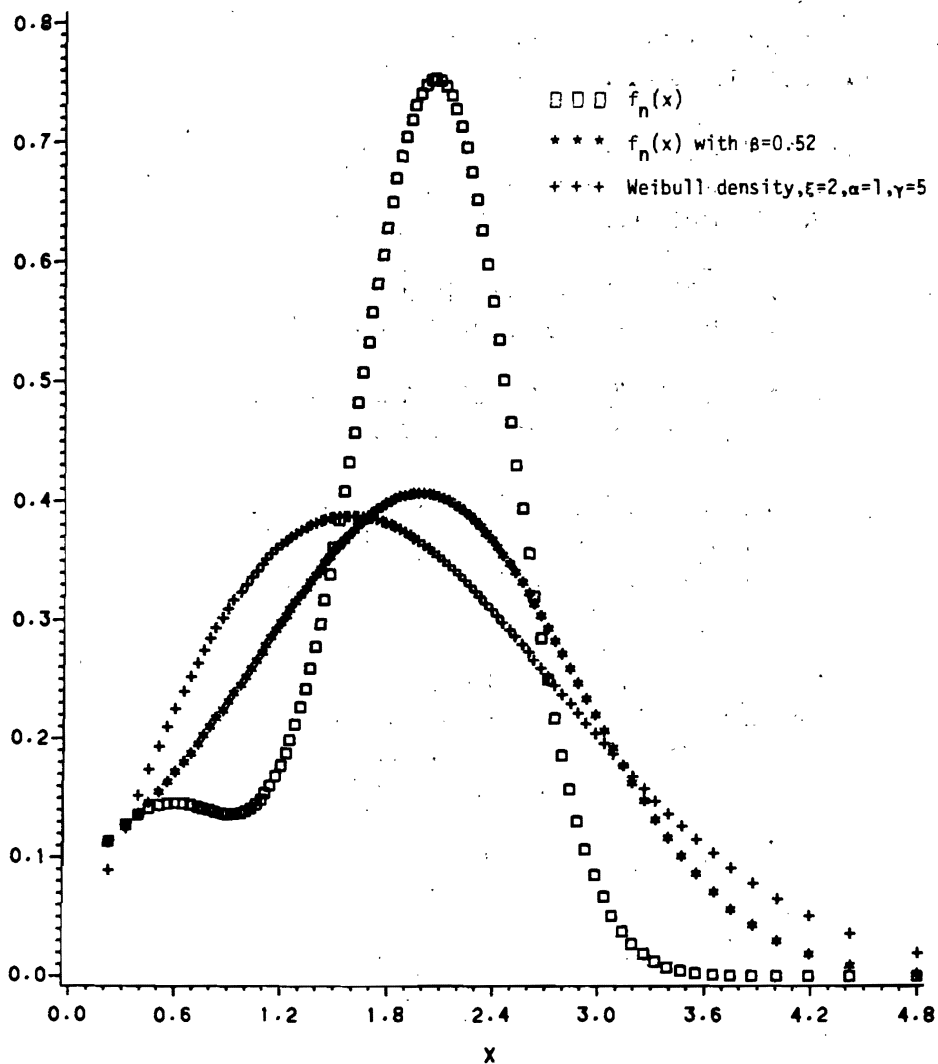


FIGURE 5. Density estimates from Weibull data for $n=60$

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